Style Haven: Shift-Left Test Pyramid Master Specification

Document Title: Style Haven - Test Pyramid Layer-by-Layer Master Specification
Platform: Fashion E-commerce Marketplace
Document Version: 3.0.0-PROD
Author: Principal QA Test Architect
Scope: Exhaustive catalog of all 33 test scenarios cleanly partitioned into dedicated sections: **Unit Tests**, **Component Tests**, **Integration & API Tests**, and **End-to-End (E2E) Tests**.

Executive Overview: Test Pyramid Architecture



SECTION 1: UNIT TESTS LAYER

Shift-Left Focus: Shifted completely to developer local environment and pre-commit hooks (`husky`).
Target Execution Speed: `< 10ms` per test (Full suite runs in `< 1s`).
Recommended Frameworks: `Vitest` / `Jest`
Scope: Pure functions, mathematical formulas, regex validators, string sanitizers, and array filtering algorithms.

Test Case ID	Req Ref	Test Scenario Title	Specific Logic / Pure Function Tested	Test Input & Conditions	Expected Outcome	Assigned Engineer
TC-CRT-004-U	Req 3.3	Promo code discount & subtotal calculation	`calculateCartTotals(items, coupon, taxRate)`	Subtotal: $95.00, Coupon: `SUMMER15` (15%), NY Tax: 8.875%, Free Shipping Threshold: $100	Discount = -$14.25; New Subtotal = $80.75; Shipping added ($10.00); Tax = $7.17; Total = $97.92	QA Eng 2
TC-CAT-004-U	Req 2.2	Size chart unit conversions (Metric vs Imperial)	`convertDimensions(val, unit)`	Imperial input: Chest = 38.0", Waist = 32.0", Hips = 40.0"	Function converts accurately using `* 2.54` : Chest = 96.52 cm, Waist = 81.28 cm, Hips = 101.60 cm	QA Eng 2
TC-OPT-002-U	Req 7.3	Loyalty rewards points valuation formula	`pointsToDiscountValue(points, rate)`	User points: 500 points, conversion rule: 100 points = $1.00	Returns exact monetary credit of `$5.00` ; negative points throw `InvalidPointException`	QA Eng 1
TC-ACC-001-U	Req 1.1	Password complexity & email format regex	`validatePasswordStrength(pwd)` , `validateEmail(email)`	Passwords: `weakpass` vs `SecurePass#2026!` ; Emails: `user@domain` vs `valid@stylehaven.io`	Weak password rejected with specific policy violation flags; valid email and 12+ char password return `isValid: true`	QA Eng 1
TC-ACC-004-U	Req 1.2	Profile address XSS input sanitizer	`sanitizeAddressInput(rawString)`	Input: `Apt #4-B, St. John's`	Script tags stripped/escaped; harmless punctuation preserved: `Apt #4-B, St. John's`	QA Eng 1
TC-SUP-002-U	Req 6.2	FAQ keyword search array filter	`filterFaqList(faqs, query)`	25 FAQ items, search query: `"refund timeline"`	Returns only 2 matching entries tagged with returns/refunds; case-insensitive	QA Eng 1

SECTION 2: COMPONENT TESTS LAYER

Shift-Left Focus: Frontend UI isolated testing; runs on every Pull Request build.
Target Execution Speed: `~200ms – 500ms` per component.
Recommended Frameworks: `Playwright Component Testing` , `React Testing Library (RTL)` , `Cypress Component`

Scope: Interactive UI states, swatches, form error styling, disabled button states, modal drawers, and accessibility (ARIA).

Test Case ID	Req Ref	Test Scenario Title	Component Under Test	Component Props / State	Expected Render & State Behavior	Assigned Engineer
TC-CAT-003-C	Req 2.2	PDP variant switcher & inventory reflection		Variant matrix with `[Cream, M]` stock = 0, `[Black, M]` stock = 5	Clicking Cream + M renders disabled "Out of Stock" button; switching to Black + M enables button with "Low Stock: Only 5 left!"	QA Eng 2
TC-CAT-004-C	Req 2.2	Size chart drawer & image zoom modal	,	`isOpen: true`, high-res image URL: `blazer_hero.webp`	Drawer renders with accessible focus trap; unit toggle switches table values; hover zooms image without pixelation	QA Eng 2
TC-CRT-003-C	Req 3.2	Cart quantity adjustment exceeding stock		`currentQty: 2`, `maxAvailableStock: 2`	Clicking `+` button is disabled; inline banner renders: *"Only 2 units available in stock"*; quantity clamped at 2	QA Eng 2
TC-CHK-001-C	Req 4.1	Shipping tier radio selection		Rates: Standard ($8.00, 3-5 days), Express ($20.00, 1-2 days)	Selecting Express radio highlights row and dynamically updates estimated delivery date banner without page reload	QA Eng 3
TC-SUP-002-C	Req 6.2	FAQ accordion expand/collapse transitions		List of 10 FAQ questions, initial state: collapsed	Clicking question expands answer with smooth CSS animation; updates `aria-expanded="true"`; closes on second click	QA Eng 1
TC-OPT-001-C	Req 7.1	Wishlist Heart icon toggle		Initial state: `isSaved: false`	Clicking button toggles SVG fill state to active red; fires micro-bounce animation; calls `onToggle(productId)` callback	QA Eng 1
TC-ACC-004-C	Req 1.2	Address form validation & error borders		Form inputs with missing postal code or invalid characters	Leaving ZIP empty highlights field with red border and renders inline error: *"Postal code is required"*	QA Eng 1

SECTION 3: INTEGRATION & API TESTS LAYER

Shift-Left Focus: Backend services, REST/GraphQL APIs, database persistence, and message brokers; runs on PR builds.
Target Execution Speed: `~50ms – 300ms` per endpoint.
Recommended Frameworks: `Supertest`, `Playwright API`, `k6` (concurrency), `Testcontainers`
Scope: Controller validation, database transactions, concurrency row-locks, token lifecycles, and security barriers.

3.1 Functional Integration & API Tests

Test Case ID	Req Ref	Test Scenario Title	Target Endpoint / Event	Test Request Payload	Expected Response & DB Assertion	Assigned Engineer
TC-ACC-001-A	Req 1.1	User registration endpoint	`POST /api/v1/auth/register`	`{ email: "valid@sh.io", password: "SecurePass#2026!" }`	HTTP 201 Created; password hashed via bcrypt in DB; verification token created in Redis/DB	QA Eng 1
TC-ACC-003-A	Req 1.1	Account lockout defense (Brute Force)	`POST /api/v1/auth/login`	5 consecutive invalid password payloads within 2 mins	6th attempt returns HTTP 429 Too Many Requests; lockout flag set in Redis for 15 mins	QA Eng 1
TC-ACC-004-A	Req 1.2	Address book persistence	`POST /api/v1/user/addresses`	`{ addressLine1: "452 Fashion Ave", isDefault: true }`	HTTP 200 OK; address persisted in DB; prior default address toggled to `isDefault: false`	QA Eng 1
TC-ACC-005-A	Req 1.3	Order history pagination	`GET /api/v1/user/orders?page=2&limit=10`	Authenticated user JWT with 25 past orders	HTTP 200 OK; payload returns exactly 10 order records with correct total page count metadata	QA Eng 1
TC-CAT-001-A	Req 2.1	Faceted product search against index	`GET /api/v1/catalog/search?category=blazers&brand=atelier&size=M`	URL query params executed against Elasticsearch cluster	HTTP 200 OK; only products matching all 3 filter criteria returned; response time < 200ms	QA Eng 2
TC-CAT-002-A	Req 2.1	Typo tolerance and fuzzy search fallback	`GET /api/v1/catalog/search?q=vntage+jaket`	Misspelled search tokens	HTTP 200 OK; fuzzy match suggests "vintage jacket"; returns relevant items; zero empty results	QA Eng 2
TC-CAT-005-A	Req 2.3	Personalized recommendation scoring	`GET /api/v1/recommendations/user`	User JWT with past viewing history: 5 outerwear items	HTTP 200 OK; returns items scored by affinity vector for category "outerwear" and designer affinity	QA Eng 2
TC-CRT-002-A	Req 3.1	Guest cart session migration upon login	`POST /api/v1/cart/merge`	Header: User JWT, Body: `{ guestCartSessionId: "uuid-123" }`	HTTP 200 OK; guest items consolidated into database user cart; duplicate SKU quantities merged	QA Eng 2

Test Case ID	Req Ref	Test Scenario Title	Target Endpoint / Event	Test Request Payload	Expected Response & DB Assertion	Assigned Engineer
TC-CRT-005-A	Req 3.3	Dynamic cart price update reconciliation	`GET /api/v1/cart`	Cart contains SKU whose catalog price dropped from $120 to $100	HTTP 200 OK; cart total reflects updated $100 price; response contains `priceChangedAlert: true`	QA Eng 2
TC-CHK-003-A	Req 4.2	Payment gateway decline error handling	`POST /api/v1/orders/checkout`	Payment payload using Stripe decline token (`card_declined`)	HTTP 402 Payment Required; non-technical error returned; order remains in `UNPAID` status	QA Eng 3
TC-CHK-005-A	Req 4.3	Asynchronous order confirmation event	Message Queue (`RabbitMQ` / `Kafka`)	Event emitted: `order.created` with `{ orderId: "#SH-98241" }`	Notification worker consumes event and dispatches email via SendGrid sandbox within 5s	QA Eng 3
TC-SEL-001-A	Req 5.1	Product listing creation with image upload	`POST /api/v1/seller/products`	Multipart form-data with variant SKUs, prices, WebP images	HTTP 201 Created; images processed to CDN; listing records written to relational catalog DB	QA Eng 2
TC-SEL-002-A	Req 5.2	Low stock automated alert trigger	`POST /api/v1/inventory/decrement`	Product inventory decremented from 3 units to 2 units	Alert daemon triggers; generates low stock badge and dispatches automated restock notification email	QA Eng 2
TC-SEL-004-A	Req 5.3	IDOR & RBAC cross-seller security	`GET /api/v1/seller/orders/999`	Seller A JWT attempting to view Order #999 owned by Seller B	HTTP 403 Forbidden or 404 Not Found; zero customer or financial data leaked	QA Eng 3
TC-SUP-001-A	Req 6.1	Real-time Live Chat WebSocket exchange	`WS /api/v1/chat/ws`	Client handshake with JWT; transmits message payload	Connection established; bidirectional message delivered to agent dashboard in < 500ms	QA Eng 1

3.2 Non-Functional, Concurrency & Security Edge Scenarios

Edge ID	Scenario Title	Target Vector	Test Methodology	Expected Result	Assigned Engineer
SEC-01	**Inventory Race Condition (Double Spend)**	Stock Locking Engine	`k6` script sends 50 concurrent `POST /checkout` requests for 1 remaining item	Exactly 1 transaction succeeds (HTTP 201); 49 requests receive HTTP 409 Conflict; stock never drops below 0	QA Eng 3
SEC-02	**Payment Webhook Loss & Network Recovery**	Stripe Webhook Listener	Drop browser connection post-3DS; send raw webhook `payment_intent.succeeded`	Backend fulfills order idempotently; order status transitions to `PAID` without browser redirect	QA Eng 3
SEC-03	**IDOR & Authorization Fuzzing**	Seller & Order Endpoints	Automated OWASP ZAP script fuzzing user/order IDs across authorization barriers	All cross-tenant access attempts return HTTP 403/404; zero PII exposed	QA Eng 3
SEC-04	**WebSocket Reconnection & Message Replay**	Live Chat Connection	Disconnect network socket for 15s; queue 2 client messages; restore connection	Socket automatically re-authenticates; flushes queued messages; reconciles conversation state	QA Eng 1
SEC-05	**XSS Injection in Reviews & Seller Notes**	Review Submission API	Submitting in product review body	Payload is HTML-encoded on ingestion; rendered harmlessly as plaintext text in browser	QA Eng 2

SECTION 4: END-TO-END (E2E) TESTS LAYER

Shift-Left Focus: Reserved strictly for high-value Golden User Journeys and cross-domain third-party interactions.
Target Execution Speed: `~45s – 2 minutes` per test flow.
Recommended Frameworks: `Playwright E2E Runner` (Chromium, WebKit, Firefox)
Execution Gate: Merge to `main` , Nightly regression runs, and Release Candidate sign-offs.

Test Case ID	Req Ref	Golden Flow Name	Complete User Journey & Interaction	Browser Assertion & Verification Points	Assigned Engineer
TC-ACC-002-E	Req 1.1	Social OAuth Login (Google & Apple)	User on login modal -> Clicks "Continue with Google" -> Browser navigates to Google consent -> User approves -> Callback to Style Haven	Browser receives OAuth callback; establishes authenticated session cookie; navbar updates to user avatar	QA Eng 1
TC-CRT-001-E	Req 3.1	Add to Cart & Mini-Cart Flyout Flow	Browse to PDP -> Select Size "M" + Color "Black" -> Click "Add to Cart"	Mini-cart drawer flies out from right side; line item reflects correct SKU and $120.00; cart badge updates from 0 to 1	QA Eng 2
TC-CHK-002-E	Req 4.2	**Master Golden Shopper Checkout**	Cart -> Proceed to Checkout -> Confirm default shipping address -> Choose Express Shipping -> Input Card -> Solve 3DS OTP iframe -> Place Order	Payment captures; browser redirects to `/order-success/{orderId}` ; order badge increments in Order History; cart resets to 0	QA Eng 3

Test Case ID	Req Ref	Golden Flow Name	Complete User Journey & Interaction	Browser Assertion & Verification Points	Assigned Engineer
TC-CHK-004-E	Req 4.2	Digital Wallet Transaction (Apple Pay / Google Pay)	User selects "Apple Pay" on checkout -> Web Payment sheet displays -> Biometric authentication completed	Order creates seamlessly without manual shipping address input; auth token validates on backend	QA Eng 3
TC-SEL-003-E	Req 5.3	Seller Order Fulfillment & Label Generation	Seller logs into dashboard -> Navigates to Unfulfilled Orders -> Inputs parcel weight -> Clicks "Generate Label"	Carrier API returns tracking number; order transitions to `SHIPPED`; printable shipping label PDF renders in browser	QA Eng 3

SECTION 5: MASTER SUMMARY & EXECUTION CADENCE

Section	Pyramid Layer	Total Test Scenarios	Proportion	Average Speed	CI/CD Pipeline Stage
Section 1	**Unit Tests**	**6**	~18%	< 1 second	Pre-commit hook / Local IDE save
Section 2	**Component Tests**	**7**	~21%	~15 seconds	Every Pull Request build
Section 3	**Integration & API Tests**	**15 (+5 Edge)**	~46%	~45 seconds	Every Pull Request build
Section 4	**End-to-End (E2E) Tests**	**5**	~15%	~2 minutes	Nightly / Production Release Gate
TOTAL	**Full Quality Stack**	**33 Scenarios**	**100%**	**< 3.5 minutes total**	Continuous Shift-Left Quality Gate